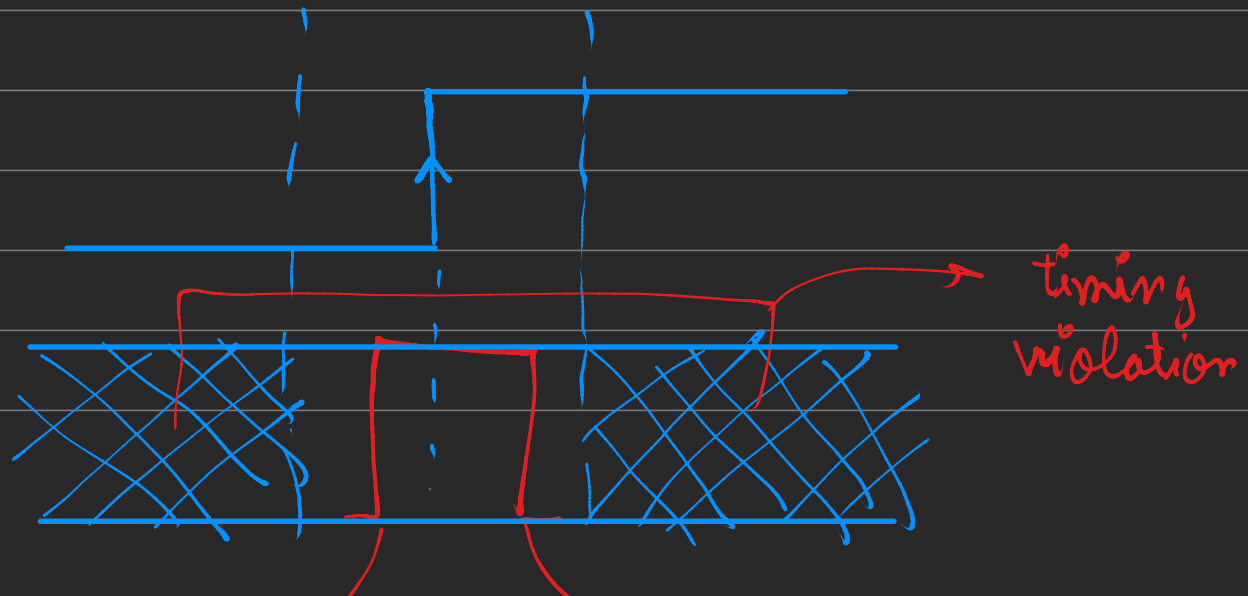


Static Timing Analysis

Setup Time: is defined as min. amount of time before the clock's active edge that the data must be stable for it to be captured correctly.

Hold Time: is after the clock's latched correctly.



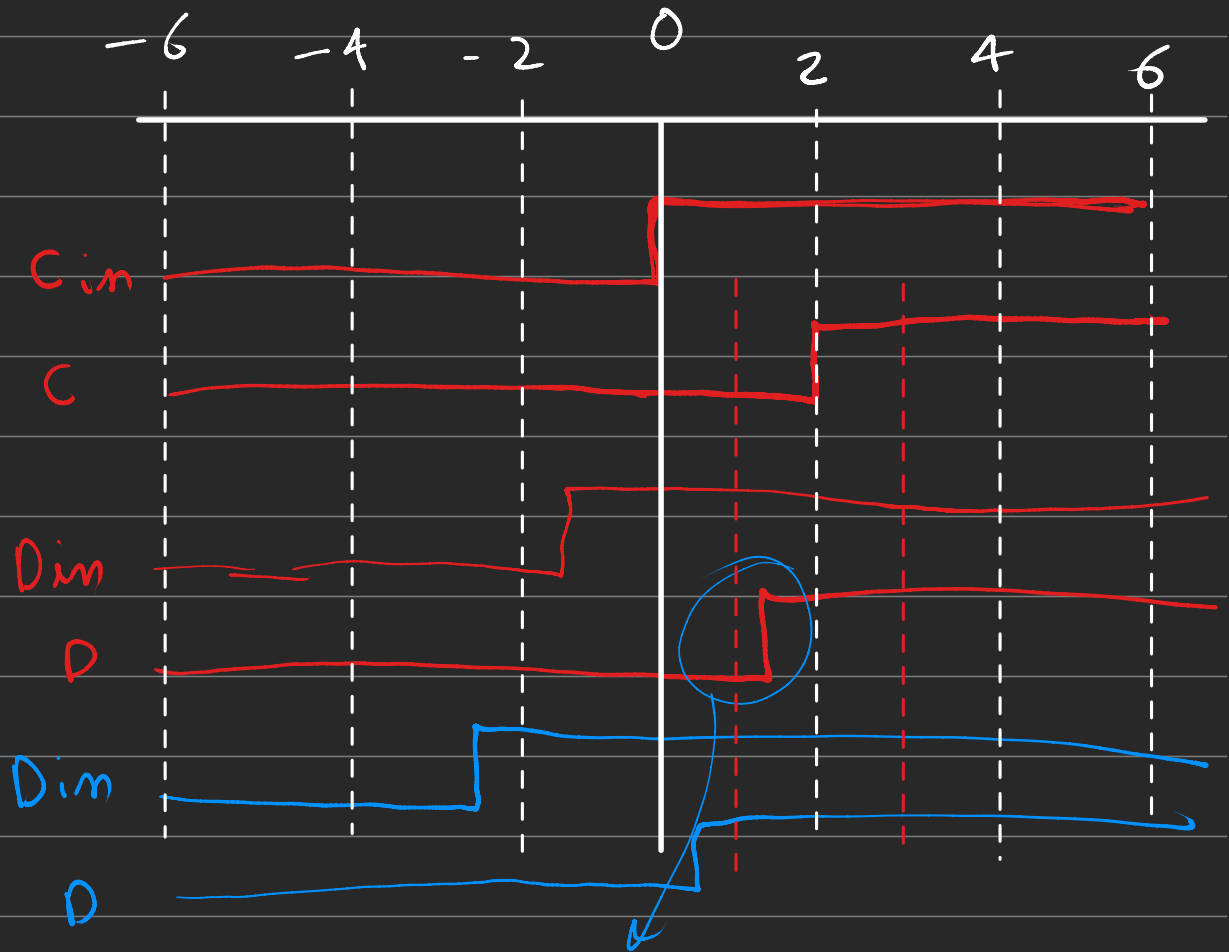
Setup violation Hold violation

EV D.F.F.

$$T_d = 3 \text{ ns}$$

$$T_c = 2 \text{ ns}$$

1 n/s time for
not violating



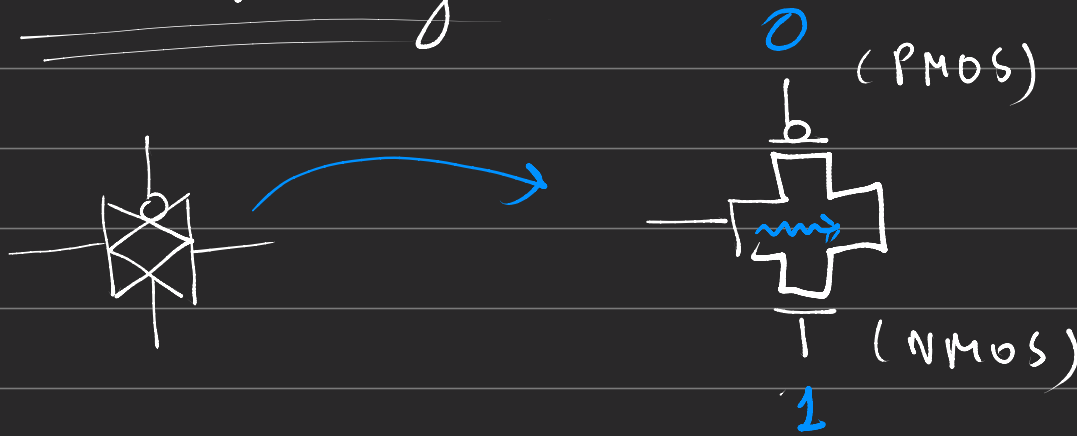
To
make
it
stable

req. for stable setup violation

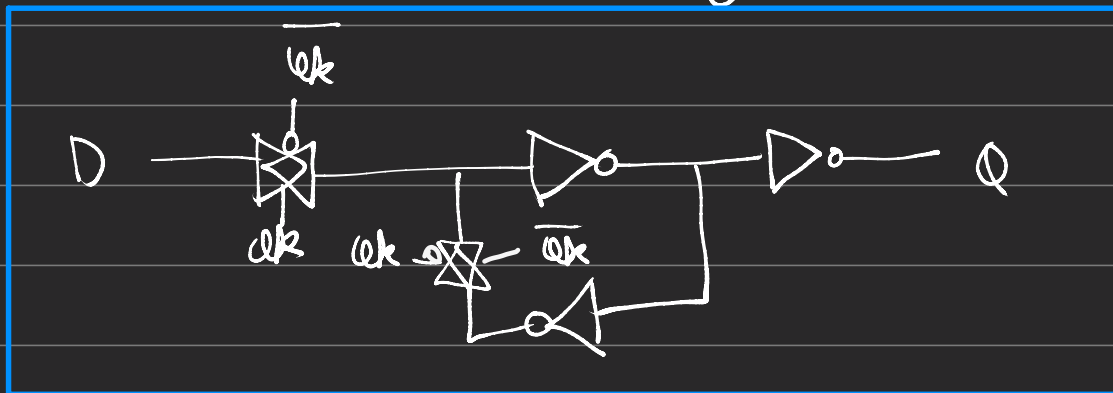
$$T_d - T_c + T_{su} \Rightarrow \text{setup time}$$

$$T_c - T_d + T_{sen} \Rightarrow \text{Hold time}$$

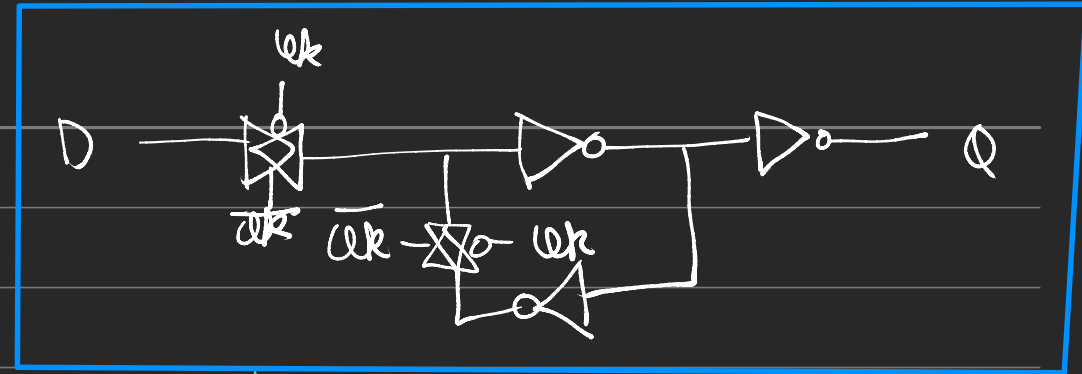
Transmission gate



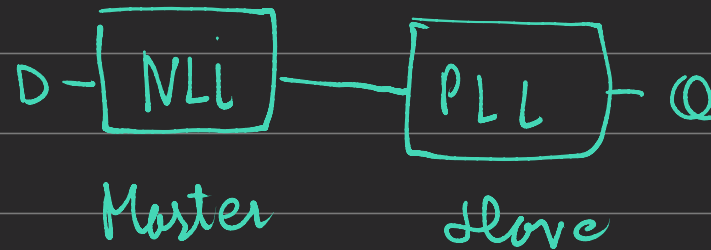
→ Internal structure of a positive level latch



→ Negative level latch



→ Positive edge latch

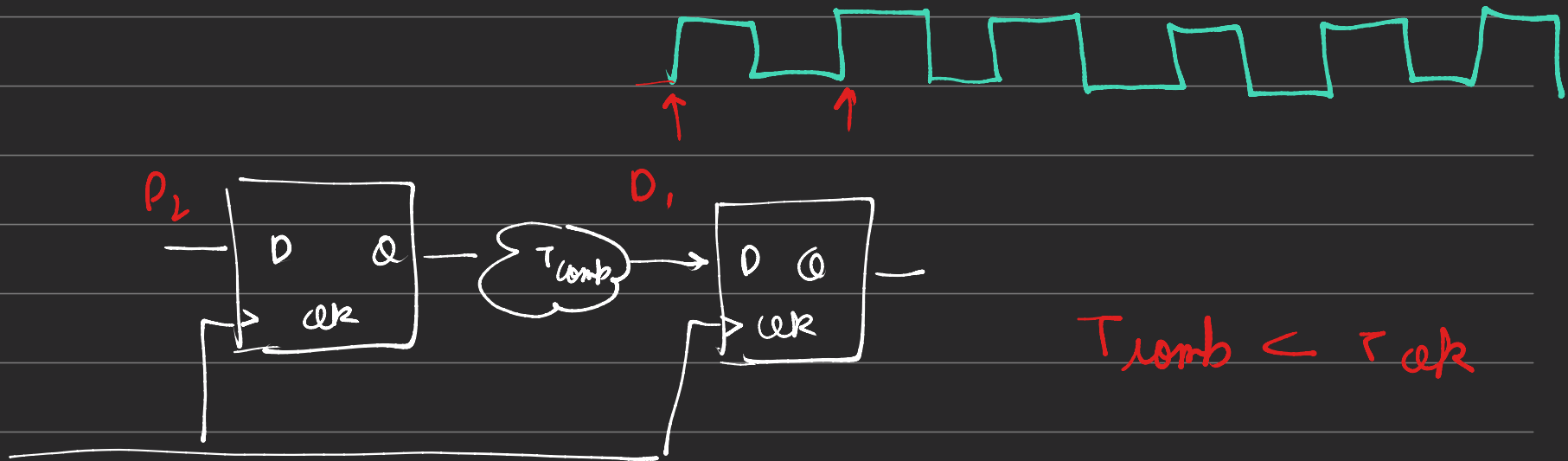


& Similarity $\ominus$ ve edge latch

PLL $\rightarrow$ NLL

Setup Analysis

①



②

If D_1 has T_{su} = 2ns → setup time

$$T_{comb} < T_{clk} - T_{su}$$

③

clock to Q delay : time after active edge to give valid

$$T_{cq} + T_{comb} < T_{clk} - T_{sen}^{output}$$

② If applied buffer at launch flop or capture flop
then

$$\Delta_1 + T_{cq} + T_{comb} < T_{clk} + \Delta_2 - T_{sen}$$

↓
Data arrival
time

↓
Data required
time.

→ It is used to determine max. clk frequency / min T_{clk} .

Holding Analysis

